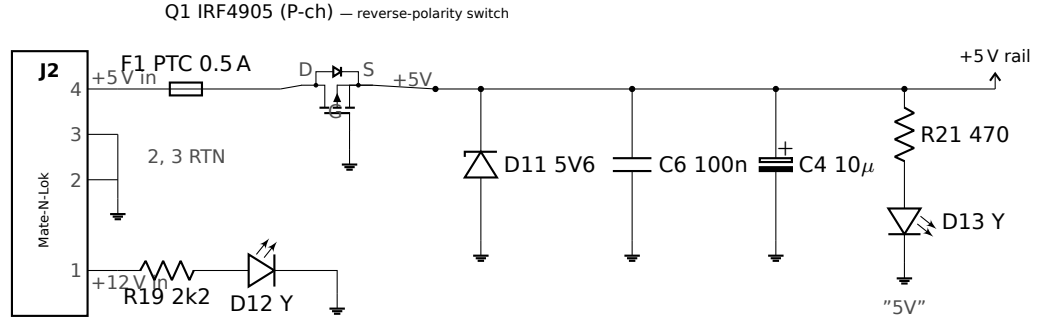
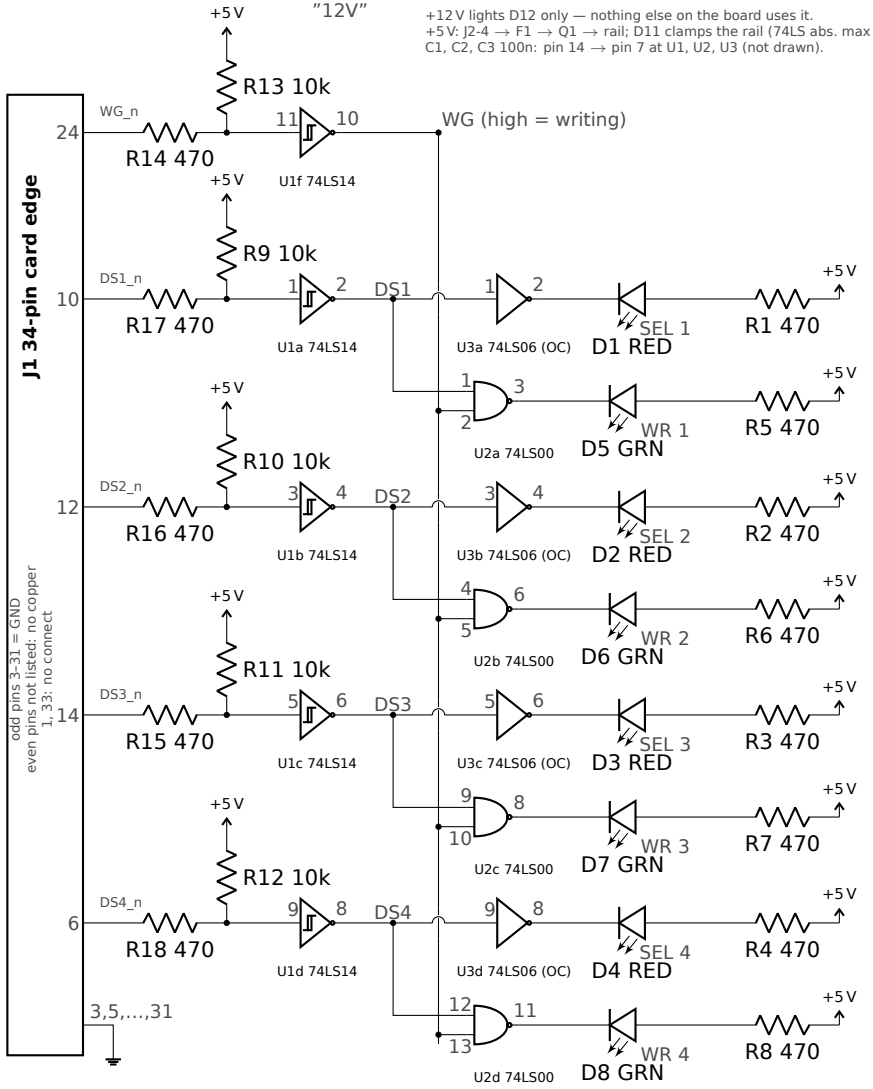
sa400_led_monitor v0.2.2 — schematic, as built (r2 BOM)

Bus signals are active-low, open-collector, terminated 150 Ω at the last drive. This board: high-impedance tap, no terminators, never drives the bus. All gate pin numbers are DIP-14 pins.



+12 V lights D12 only — nothing else on the board uses it.
+5 V: J2-4 → F1 → Q1 → rail; D11 clamps the rail (74LS abs. max 7 V).
C1, C2, C3 100n: pin 14 → pin 7 at U1, U2, U3 (not drawn).



Unused: U1 pin 13 → GND (out 12 open); U3 pins 11, 13 → GND (outs 10, 12 open). Power: pin 14 = +5 V, pin 7 = GND on U1–U3.
Drive n on this board = TRS-80 drive $: n-1$. LED current $\approx (5 - 0.4 - V_f)/470 \approx 5.5$–$6.5$ mA. Red = DSn; green = DSn · WG.